Accelerating metabolic engineering through multimodal phenotype prediction with TorchCell and the Cell Graph Transformer

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Abstract

Deep learning has advanced protein structure prediction, yet metabolic engineering lags because phenotype data are scattered across heterogeneous studies lacking a shared ontology. We present TorchCell, an open-source framework that annotates literature and public datasets with a shared ontology and ingests them into a Neo4j knowledge graph, then builds deep-learning-ready datasets in which each strain is a perturbation operator over a shared wildtype reference cell combining genome sequence, gene networks, metabolism, and environment. We develop the Cell Graph Transformer (CGT), a virtual-cell architecture in which biological interaction graphs constrain multi-head attention and an equivariant perturbation operator feeds multitask decoders. In *Saccharomyces cerevisiae*, CGT predicts trigenic gene-gene interactions (Pearson $\boldsymbol{r} = \mathbf{0.454}$), outperforming DANGO, DCell, and Yeast9 flux balance analysis, and generalizes to knockout expression ($\boldsymbol{r} = \mathbf{0.543}$) and morphology ($\boldsymbol{r} = \mathbf{0.619}$). CGT recommends gene deletions for beta-carotene and betaxanthin production, pairing with autonomous biofoundries for AI-guided strain engineering.

Keywords: metabolic engineering, foundation models, gene interactions, virtual cell, knowledge graph, strain design

1 Introduction

[FILLER – replace with Intro prose.] Deep learning transformed protein structure and function prediction, yet systems biology and metabolic engineering lag because phenotype data are scattered across small, heterogeneous studies with no shared schema [1]. Prior approaches fall short: mechanistic flux balance analysis misses epistasis, and recent perturbation transformers do not consistently beat linear baselines [2]. Our own classical baselines predict knockout fitness but fail on gene interactions (Suppl. Fig. S1), motivating a deep, biologically grounded model. Here we present TorchCell and the Cell Graph Transformer (CGT), whose data model and architecture we outline in Fig. 1.

2 Results

A shared ontology and knowledge graph unify metabolic-engineering data. *[FILLER – R1.]* TorchCell annotates literature and public datasets with a shared experiment ontology and ingests them into a Neo4j knowledge graph; dataloaders emit deep-learning-ready datasets in which each strain is a perturbation operator over a shared wildtype reference cell (Fig. 1; data scale in Suppl. Fig. S2; ontology/KG spec in Suppl. Fig. S4).

The Cell Graph Transformer: a graph-constrained virtual cell. *[FILLER – R2.]* Biological interaction graphs constrain multi-head attention through a loss aligning attention to network priors; an equivariant perturbation operator maps the reference embedding to the perturbed strain state (Fig. 2; ablations in Suppl. Fig. S3).

CGT predicts trigenic gene-gene interactions at state of the art. *[FILLER – R3.]* Classical ML fails on this task (Suppl. Fig. S1); CGT reaches Pearson $r = 0.454$, outperforming DANGO, DCell, and Yeast9 FBA (Fig. 3).

One embedding, many phenotypes: expression and morphology. *[FILLER – R4.]* The same architecture predicts knockout expression ($r = 0.543$) and single-knockout morphology ($r = 0.619$) (Fig. 4; expression diagnostics in Suppl. Fig. S5).

CGT-guided strain design with an autonomous foundry. *[FILLER – R5.]* We recommend gene deletions for beta-carotene and betaxanthin production and pair with the UIUC iBioFoundry for iterative design (Fig. 5; inference at scale in Suppl. Fig. S6).

3 Discussion

[FILLER – Discussion.] A strain-aware, biologically grounded representation is the differentiator; limitations include single-organism scope, morphology-data maturity, and error-bar provenance.

4 Methods

[FILLER – Methods.] Ontology and knowledge-graph construction; dataset and perturbation-operator formalism; CGT architecture and the graph-regularized attention loss; training, splits, metrics; baselines (DANGO, DCell, Yeast9 FBA); inference at scale; statistics and error-bar definitions.

Supplementary information.

Acknowledgments.

Declarations

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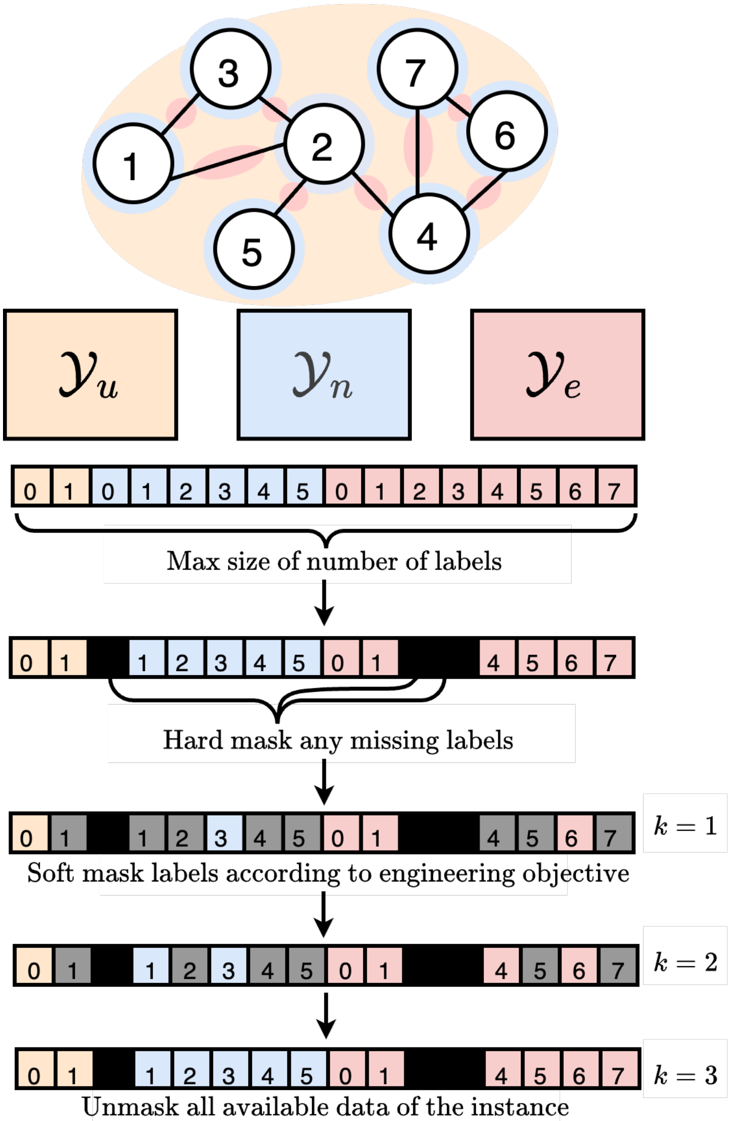

Fig. 1 TorchCell: ontology-unified knowledge graph and the perturbation-operator cell representation. Panels (a)–(d).

[Fig. 2 placeholder – compose in draw.io]

CGT architecture. (a) graph-regularized attention; (b) attention-prior alignment; (c) equivariant perturbation operator; (d) multitask decoder heads.

Fig. 2 The Cell Graph Transformer architecture. (a)–(d).

[Fig. 3 placeholder – compose in draw.io]

Trigenic GGI. (a) model comparison (CGT/DANGO/DCell/Yeast9); (b) predicted-vs-actual; (c) performance vs interaction magnitude; (d) ablation.

Fig. 3 State-of-the-art trigenic gene-gene interaction prediction. (a)–(d).

[Fig. 4 placeholder – compose in draw.io]

Multitask generalization. (a) expression predicted-vs-actual; (b) morphology predicted-vs-actual; (c) cross-phenotype summary; (d) shared-latent UMAP.

Fig. 4 One latent cell embedding generalizes across phenotype classes. (a)–(d).

[Fig. 5 placeholder – compose in draw.io]

CGT-guided strain design. (a) beta-carotene/betaxanthin pathways; (b) ranked recommendations; (c) TorchCell + iBioFoundry DBTL loop; (d) validation titers.

Fig. 5 CGT-guided strain design closes the DBTL loop with an autonomous foundry. (a)–(d).